AI in Health Hackathon

THEME: Innovating Healthcare with Artificial Intelligence

A multidisciplinary challenge bringing together clinicians, data scientists, engineers, public health professionals, and students to build the next generation of AI-powered healthcare solutions.

Challenge Tracks

Five Tracks.
One Mission: Better Health.

Each track is designed to tackle a distinct, pressing problem in modern healthcare. Teams will select the track that best aligns with their skills and passion, then build an AI-driven solution with real-world impact.

🔬 Disease Diagnosis

Early detection models for cancer, malaria, pneumonia, and more.

🩻 Medical Imaging

AI for X-ray, MRI, CT scans, ECG, and EEG signal processing.

📊 Health Analytics

Transforming raw health data into actionable clinical decisions.

🌍 Public Health

Population-level surveillance, outbreak prediction, and dashboards.

⚖️ AI Ethics

Bias detection, explainability, and responsible AI in healthcare.

Track 1: Disease Diagnosis

This track challenges teams to build AI models that support early detection and diagnosis — helping clinicians make faster, more accurate decisions and ultimately saving lives.

Cancer Detection

Classify breast, cervical, or other cancer types from pathology images, tissue scans, or clinical data using machine learning or deep learning.

Malaria Identification

Detect malaria parasites from microscopy blood smear images using computer vision — a critical need in low-resource clinical settings.

Pneumonia Detection

Build classification systems from chest X-rays to identify pneumonia patterns, reducing radiologist workload and diagnostic delays.

Track 2: Medical Imaging & Signal Processing

This track focuses on extracting meaningful clinical insights from the rich, complex world of medical images and physiological signals — unlocking patterns that are invisible to the human eye.

What You'll Work With

- X-ray, MRI, and CT scan images
- ECG and EEG physiological signals
- Segmentation of tumors and organs
- Radiology automation pipelines

Key Technical Areas

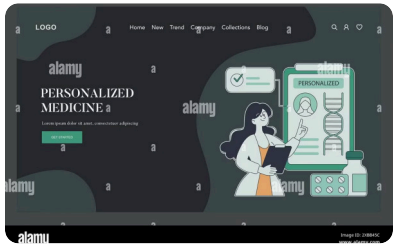
Teams should focus on **image segmentation and image classification** — precisely delineating anatomical structures or lesions — and **signal processing** for detecting cardiac arrhythmias or neurological anomalies. The goal is to develop models that meaningfully enhance clinical insight and automate repetitive interpretation tasks, freeing clinicians to focus on patient care.

Track 3: Health Data Analytics

Healthcare systems generate enormous volumes of structured and unstructured data. This track empowers teams to turn that data into decisions — improving patient outcomes, personalizing care, and making healthcare systems smarter and more efficient.

Predictive Analytics

Build models that forecast disease risk, patient deterioration, or readmission likelihood — enabling proactive interventions.



Patient Stratification

Use AI to segment patient populations and recommend personalized treatment pathways tailored to individual health profiles.

Resource Optimization

Model hospital capacity — bed allocation, staffing schedules, equipment usage — to reduce inefficiency and improve care delivery.

Track 4: Public Health Surveillance

AI has the power to transform how governments and health organizations detect, monitor, and respond to population-level health threats — moving from reactive crisis management to proactive planning.

1

Detect

Build outbreak prediction models and early warning systems that identify disease signals before they become crises.

2

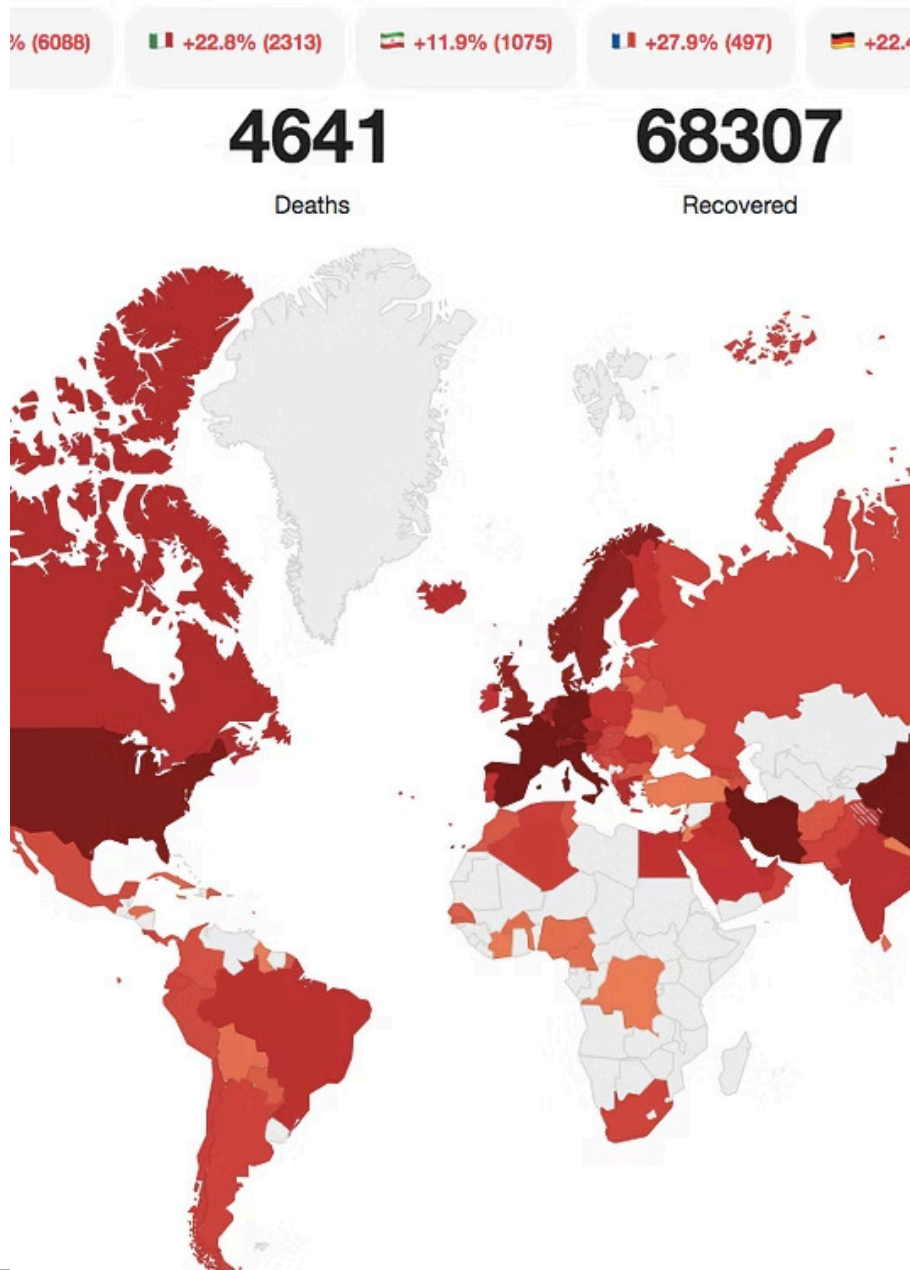
Monitor

Design real-time health monitoring dashboards that aggregate and visualize epidemiological data across regions.

3

Respond

Support policymakers with AI-powered trend analysis and scenario modeling to guide timely, evidence-based interventions.



Track 5: AI Ethics & Explainability

Building powerful AI is only half the mission. This track ensures that AI systems in healthcare are trustworthy, transparent, and equitable — because a model that clinicians and patients can't trust will never reach its full potential.

Bias Detection & Mitigation

Identify and correct algorithmic bias across demographic groups — race, gender, age — to ensure equitable AI-driven care for all patients.

Explainable AI (XAI)

Design models that clinicians can interpret and explain. Use tools like SHAP, LIME, or attention maps to make AI decisions transparent and auditable.

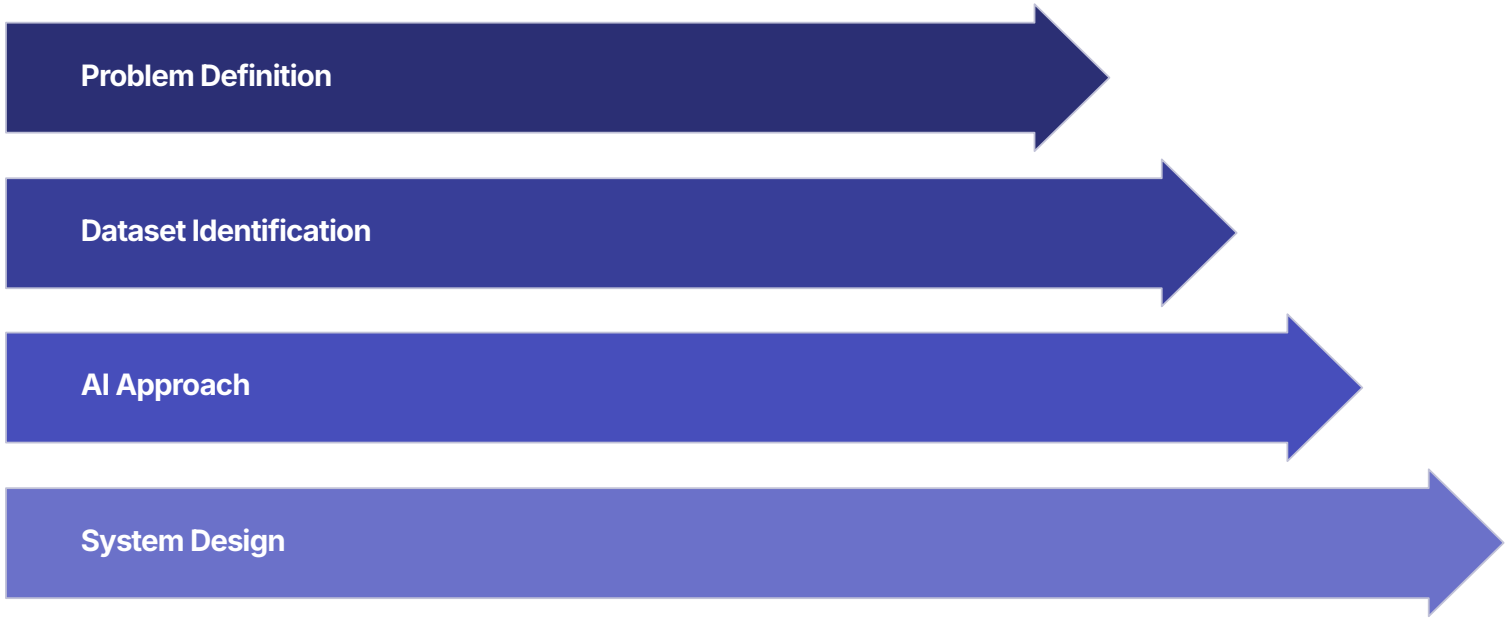
Fairness & Accountability

Ensure solutions align with healthcare regulations, patient rights, and societal values — building systems that earn trust across institutions and communities.

Team Deliverables

Your Structured Approach to Building a Solution

Every team follows a structured development process — from defining the problem to designing a complete AI system. This framework ensures your solution is rigorous, reproducible, and ready to present.



Each phase builds on the last, guiding your team from a clearly scoped healthcare challenge to a functional prototype backed by sound methodology and ethical awareness.

Step 1: Problem Definition

The foundation of every great solution is a well-defined problem. Before writing a single line of code, teams must deeply understand the healthcare challenge they are addressing and why it matters.

What to Define

- The specific healthcare problem and its clinical or social context
- Target users: clinicians, patients, public health officials, or policymakers
- Real-world significance and measurable impact
- Gaps or limitations in existing solutions

What Makes It Strong

A well-defined problem is **specific** — not "improve healthcare" but "reduce diagnostic delays for cervical cancer in low-resource clinics." It is **relevant** to a real user need and **impactful** enough to justify an AI-powered solution. Judges will evaluate how clearly and compellingly you frame the problem before assessing your technical approach.

Step 2: Dataset Identification & Justification

AI models are only as good as the data they learn from. Teams must be thoughtful and transparent about their data choices — addressing quality, ethics, and relevance from the outset.

1

Source Your Data

Use provided datasets or locate relevant public repositories (e.g., NIH, PhysioNet, Kaggle Health). Justify why this dataset is appropriate for your problem.

2

Describe Characteristics

Document dataset size, data type (images, tabular, text), key features, modality, and any known biases or limitations in the data.

3

Address Data Quality

Identify and address missing values, class imbalance, noise, and inconsistencies. Propose augmentation or simulation strategies if data is limited.

4

Consider Ethics

Address patient privacy, informed consent, data anonymization, and security. Responsible data use is a core evaluation criterion.

Step 3: Designing Your AI Approach

With your problem defined and your data understood, it's time to select and justify the AI methodology that best fits your challenge. Innovation is encouraged — but feasibility matters equally.

Model Selection

Choose the right architecture: classical ML (Random Forest, SVM), CNNs for imaging, RNNs or Transformers for sequential or text data.

Feature Engineering

Define your feature extraction or representation strategy — from hand-crafted features to end-to-end learned representations.

Training Strategy

Outline your training pipeline: data splits, cross-validation, hyperparameter tuning, and handling of imbalanced classes.

Evaluation Metrics

Select clinically meaningful metrics: accuracy, precision, recall, F1-score, AUC-ROC — and explain what they mean in your healthcare context.

Step 4: System Design & Architecture

A strong AI solution is more than a model — it's a complete, deployable system. Teams should design a full pipeline that could realistically be integrated into a clinical or public health setting.

1

Data Input & Preprocessing

Define how raw data enters the system and how it is cleaned, normalized, and prepared for the model.

2

Model Training & Inference

Describe how your model is trained, validated, and how predictions are generated in real-time or batch settings.

3

User Interface & Visualization

Design an intuitive front-end — a dashboard, mobile app, or web interface — that makes outputs accessible to end users.

4

Deployment Considerations

Address how the system would be deployed: web-based, mobile, cloud, or on-premise. Consider scalability, latency, and accessibility.

💡 **Pro Tip:** Block diagrams and system architecture charts are **highly encouraged.** A clear visual of your solution pipeline often communicates more than paragraphs of text — and judges will appreciate it.

Step 5: Mentor Walkthroughs & Iteration

Hackathons are not just about the final product — the process of learning, iterating, and improving matters just as much. Mentor sessions are a critical resource, and active engagement is expected.

During Mentor Sessions

- Walk mentors through your current solution design
- Seek feedback on technical feasibility and clinical relevance
- Ask for guidance on model selection, data strategy, or ethical considerations
- Come prepared with specific questions — not just a status update

The Iteration Mindset

Great solutions rarely emerge fully formed. Use mentor feedback to **pivot when necessary**, simplify where complexity isn't adding value, and sharpen the clarity of your problem framing. Teams that demonstrate thoughtful iteration — showing how their solution evolved — are evaluated more favorably than those who present a static first draft.

Final Outputs

What Every Team Must Deliver

By the close of the hackathon, each team presents a complete, cohesive solution package. These deliverables demonstrate the full arc of your work — from problem identification to real-world readiness.

Problem Statement & Solution Overview

A clear, concise description of the healthcare challenge you addressed and how your AI solution directly solves it.

Working Prototype or Proof-of-Concept

A functional demo, model output, or proof-of-concept showing that your solution works — even if not production-ready.

Technical Presentation

A polished slide deck or live demo that communicates your methodology, results, and design choices to a multidisciplinary audience.

System Architecture Diagram

A visual representation of your complete solution pipeline — data flow, components, interfaces, and deployment considerations.

Impact, Scalability & Ethics Discussion

A thoughtful reflection on the real-world impact of your solution, how it could scale, and how it addresses ethical considerations.

Build Something That Matters

This hackathon is more than a competition — it's an opportunity to apply your expertise to problems that affect millions of lives. Whether you're a clinician with deep domain knowledge, an engineer with a sharp technical mind, or a student eager to learn, your contribution is essential.

🩺 Clinicians

Bring domain expertise. Define problems that matter in real clinical practice and validate that solutions are meaningful.

💻 Engineers & Data Scientists

Build, train, and deploy. Turn healthcare challenges into working AI systems backed by sound methodology.

🌍 Public Health Professionals

Frame population-level impact. Ensure solutions address systemic gaps and can reach the communities that need them most.

🎓 Students

Learn by doing. Collaborate across disciplines, push creative boundaries, and emerge with a portfolio-worthy project.

Awards & Post-Hackathon Support

Recognizing excellence, fueling long-term impact — here's what awaits the teams who bring their boldest ideas to the hackathon.

1	2	3
Awards	**Incubation**	**Vision**
Cash prizes for the top three teams	**Continued support through the AI Center**	A pipeline of AI-driven healthcare innovation

🏆 Top 3 Teams Win Cash Prizes

Prize amounts will be officially revealed at the start of the hackathon — adding a thrilling element of anticipation and motivation for all participants from day one.

📋 Judging Criteria

- **Innovation** — originality and creativity of the solution
- **Impact** — potential benefit to healthcare outcomes
- **Technical Quality** — robustness and implementation
- **Feasibility** — realistic path to deployment
- **Ethical Considerations** — responsible AI development

Incubation & Continued Support

The hackathon is just the beginning. Promising teams — including but not limited to the winners — will be considered for **incubation through the AI Center**, transforming bold ideas into scalable, real-world healthcare solutions.

Technical Mentorship

Dedicated guidance from experienced AI and healthcare technology experts to help teams navigate complex challenges and refine their solutions.

AI Infrastructure Access

Selected teams gain access to cutting-edge AI infrastructure, computing resources, and datasets essential for building and testing deployable prototypes.

Research & Publication

Opportunities to co-author research publications and collaborate with academic and clinical partners, elevating the credibility and reach of your work.

Industry Linkages

Strategic connections to healthcare institutions, industry partners, and potential funders to accelerate the path from prototype to real-world impact.

The incubation program is designed to transform hackathon ideas into **scalable, sustainable, and impactful healthcare solutions** — giving every promising team a genuine runway beyond competition day.